Warcraft Card Duels

Technical & Business Analysis

ASE 330 Project

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1. Golden Pitch Canvas

Why

Strategy/RPG players want meaningful decisions, character variety, and engaging turn-based combat without a complicated system or a massive time commitment.

How

Three-hero teams · unique classes and abilities · turn-based combat · random encounters · dungeon progression · bosses

What

A Python/Pygame turn-based RPG where players build a team of three heroes and fight through dungeons toward powerful boss encounters.

2. Customer Personas

The Strategy Player

- Enjoys turn-based games and meaningful decisions
- Goals: build an effective team, experiment with abilities, beat harder encounters
- Pain points: complex systems, repetitive combat, decisions with little impact

The RPG Character Player

- Enjoys classes, abilities, and progression
- Goals: discover heroes, try specializations, build teams around strengths
- Pain points: small rosters, characters that feel too similar, limited customization

3. Value Proposition: Strategy Player

Jobs	Build a team, choose abilities, react to enemy groups, beat harder fights
Pains	Repetitive battles, predictable encounters, no meaningful choices
Gains	Strategic choices, varied team combos, challenge
Products	3-hero teams, turn-based combat, unique abilities, random encounters, bosses
Pain relievers	Different abilities drive different strategies; randomized encounters and enemy types vary every battle
Gain creators	Team composition and specializations create roles; difficulty rewards good decisions

3. Value Proposition: RPG Character Player

Jobs	Discover heroes, build around strengths, learn abilities, progress through dungeons
Pains	Characters that lack differences, limited roster, repetitive progression
Gains	Diverse roster, unique classes, distinct abilities, many team combos
Products	Planned hero roster, classes/specializations, stats, 3–5 abilities each, team selection
Pain relievers	Heroes have distinct strengths; choose any three; encounters push experimentation
Gain creators	A diverse roster means more combinations; unique abilities make heroes feel distinct

4. Target Market (TAM / SAM / SOM)

- **TAM – Global PC gaming:** ~$43.6B revenue, ~936M players (Newzoo, 2025)
- **SAM – PC RPG market:** ~$20.64B (~21.9% of a $94.2B global RPG market; PW Consulting). Global turn-based RPG segment: ~$14.01B (context only)
- **SOM – Realistic for solo developer game:** a small, dedicated group of players, mainly friends and those in the space

5. Strengths & Weaknesses

Strengths	Weaknesses
Team-based turn-based combat	Small team, limited time and resources
Distinct hero abilities and roles	Less polished graphics
Modular, easy-to-expand architecture	Smaller roster and content than established RPGs

5. Opportunities & Threats

Opportunities	Threats
Growing interest in turn-based RPGs	Many competing RPGs and big franchises
Expandable heroes, abilities, dungeons	Player attention is hard to capture
Possible Steam release; community feedback	Far larger commercial budgets

5. PESTEL Analysis

- **Economic:** Crowded indie/RPG market; no marketing budget, so success depends on word of mouth and retention; budget is mostly development time
- **Social:** Players want variety, depth, and visible progression (Mom Tests); turn-based RPGs keep a dedicated audience
- **Technological:** Python/Pygame is accessible and runs on modest hardware; modular design eases expansion; no high-end graphics
- **Legal:** Must avoid copyrighted Warcraft assets; respect library licenses; follow platform rules if sold commercially

Political and Environmental omitted: no material impact on a solo, lightweight PC game.

6. Competitor Analysis

Game	Team	Turns	Variety	Main difference
World of Warcraft	Yes	No	High	MMO-scale, real-time
Darkest Dungeon	Yes	Yes	High	Complex stress/survival systems
Clair Obscur: Expedition 33	Yes	Yes	High	Larger cinematic RPG
Warcraft Card Duels	**3 heroes**	**Yes**	**Yes**	**Focused, accessible team strategy**

Our advantage is focus: a smaller, accessible turn-based RPG centered on three-hero team building and strategic ability choices, not competing on graphics, or production value.

7. Mom Test Findings

Problems: repetitive, predictable combat · decisions that don't matter · little meaningful progression · lack of strategic depth

Alternatives used: World of Warcraft, Baldur's Gate 3, Clair Obscur: Expedition 33, Pokémon

Desired: clear progression goals · strategic combat · team synergy · distinct characters · varied encounters · engaging bosses · rewards and achievements · goals beyond just playing

Behavior: Most respondents play the genre for hours daily or several times a week, but some quit when content felt boring or progression felt too slow.

7. Mom Test: Product Implications

- **Hero variety:** keep a diverse roster; 30 may be too many to keep unique
- **Team size:** keep 3 – variety without overload
- **Abilities:** need synergy and counterplay, not reskins
- **Combat:** enough depth to require planning, but short and simple
- **Encounters:** unique enemy behavior – a top priority
- **Progression:** needs the most attention – rewards, achievements, collections, stronger abilities

User Experience & Retention

Gap: nothing yet gives players a reason to come back.

Proposed improvements

- **Daily Dungeon:** one fixed-seed run per day with a special modifier
- **Daily Reward:** first win of the day earns a bonus
- **Hero Unlocks:** earn heroes over time to create a collection goal
- **Achievements & Streaks:** e.g., beat a boss with a mono-class team
- **Quick Battles:** clear turn order and ability tooltips

Unlocks and achievements first; daily features if time allows.

Business Model Canvas: Partners, Activities & Resources

Key Partners

- Playtesters and Mom Test participants who give feedback during development
- Classmates and instructor who review progress
- Artists or audio creators, only if assets are commissioned or licensed

Key Activities

- Build the combat and encounter loop
- Design, balance, and expand the hero roster
- Add retention features (unlocks, achievements, daily rewards)
- Playtest and refine from user feedback

Business Model Canvas: Value & Customers

Value Propositions

For the strategy player

- Meaningful decisions in every battle
- Different team combinations and strategies
- Varied encounters, enemy behaviors, and bosses
- Challenge that rewards good planning

For the RPG character player

- Diverse roster with distinct classes and abilities
- Freedom to build a team of three around favorite heroes
- Unlocks, achievements, and daily rewards to keep progressing

Business Model Canvas: Relationships & Channels

Customer Relationships

- Self-service and easy to learn
- Community feedback shapes new content
- Goals and rewards (achievements, daily dungeon, streaks)
- Regular updates with new heroes and dungeons

Channels

- Steam or itch.io (potential release platforms)
- GitHub repository
- Word of mouth
- RPG communities and social media

Business Model Canvas: Costs, Revenue & Metrics

Cost Structure

- Developer time (largest cost)
- Art, audio, and asset creation or licenses
- Platform fees if released commercially
- Testing and marketing

Revenue Streams

- One-time game purchase (~$5 planning price)
- Possible future paid expansions (hero or dungeon packs)

Key Metrics

- *Daily active players:* unique players who launch the game each day

8. Feasibility Plan

Prototype: Python + Pygame + Modular Structure (uv for environment, unit tests for heroes and enemies).

Phase	Status	Scope
1. Foundation	**Done**	Game loop, states, menus, heroes, enemies, abilities, team and dungeon selection, 1–3 enemies per encounter
2. Combat	Next	Turn order, actions, targets, damage, enemy turns, win/lose
3. Progression	Planned	Enemy behaviors, difficulty scaling, bosses, full gameplay loop
4. Content & Polish	Planned	More heroes and abilities, balancing, UI, assets, retention features, testing

8. Feasibility Assessment

Area	Feasibility
Python / Pygame, hero and team system	High
Turn-based combat, encounters, enemy behaviors, bosses	High
Core gameplay loop	High
15–30 heroes	Moderate
High-end graphics	Low
Multiplayer	Outside scope

Focused single-player scope plus an existing, expandable foundation makes the project feasible.

Conclusion

Warcraft Card Duels delivers a simple, strategic, replayable RPG where **team composition and ability choices decide each encounter**.

- Foundation already built: states, screens, heroes, enemies, abilities, selection
- Next: finish combat and encounters, add retention features, then expand content and test with users